

Official-source Electricity compilation 06

Mixed-source chapter worksheet • cross-paper reuse disclosed

Total: 33 marks • Suggested time: 66 minutes

Instructions and reuse disclosure

Curated compilation of official CBSE source questions, not a CBSE-issued chapter paper. Original source labels and exact source images are retained.

Answer every selected question. Only one selected source-slot alternative appears in this paper; do not answer or count excluded alternatives.

Questions recur across this 15-paper collection. Reuse is disclosed per question in the source appendix; these are not 15 disjoint official papers.

Sections use available source marks: A MCQ; B 2 marks; C 3 marks; D 5 marks or source-marked case questions. Case marks are not inflated to five.

Section A — MCQ

Compilation questions 1-4; 4 marks. Exact source labels remain attached to each question.

Section B — 2-mark questions

Compilation questions 5-8; 8 marks. Exact source labels remain attached to each question.

Section C — 3-mark questions

Compilation questions 9-12; 12 marks. Exact source labels remain attached to each question.

Section D — 5-mark / case-based questions

Compilation questions 13-14; 9 marks. Exact source labels remain attached to each question.

Section A — MCQ

1. [1 marks] Source Q14 • main

14. An electric iron of resistance $20\ \Omega$ draws a current of 5 A. The heat developed in the iron in 30 seconds is :

- | | |
|-------------|------------|
| (A) 15000 J | (B) 6000 J |
| (C) 1500 J | (D) 3000 J |

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

2. [1 marks] Source Q6 • 2019-20

- 6 When a 4V battery is connected across an unknown resistor there is a current of 100 mA in the circuit. The value of the resistance of the resistor is:
- i) $4\ \Omega$
 - ii) $40\ \Omega$
 - iii) $400\ \Omega$
 - iv) $0.4\ \Omega$

Source/editorial note: Original spelling “resister” retained. Q6 mark is printed on the preceding page (page 3); Section A instructions and the MS both confirm 1 mark.

3. [1 marks] Source Q13 • 2022-23

13	A complete circuit is left on for several minutes, causing the connecting copper wire to become hot. As the temperature of the wire increases, the electrical resistance of the wire	1
	(a) decreases.	
	(b) remains the same.	
	(c) increases.	
	(d) increases for some time and then decreases .	

4. [1 marks] Source Q15 • compartment

15. The resistance of a wire does **not** depend on its :
- (a) Length
 - (b) Area of cross-section
 - (c) Shape
 - (d) Material

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

Section B — 2-mark questions

5. [2 marks] Source Q25 main • 2023-24

25	A piece of wire of resistance R is cut into three equal parts. These parts are then connected in parallel. If the equivalent resistance of this parallel combination is R_1 , what is the value of the ratio $R_1 : R$?	2
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Source/editorial note: Main alternative only. OR concerns magnetic field patterns and is excluded. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

6. [2 marks] Source Q34 VI A • 2025-26

For visually impaired students

- A. You are given four resistors each having resistance of R ohm. Find the maximum and minimum resistance that can be made with these four resistors.

Source/editorial note: Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

7. [2 marks] Source Q34 B • 2026-27

Or

- B.
I. Define resistivity.
II. Justify with an example why resistivity is an intrinsic property of a material.

Source/editorial note: Original MS “thick-nesses” retained; comparison implicitly at fixed temperature. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

8. [2 marks] Source Q25 VI A • 2024-25

For visually impaired students

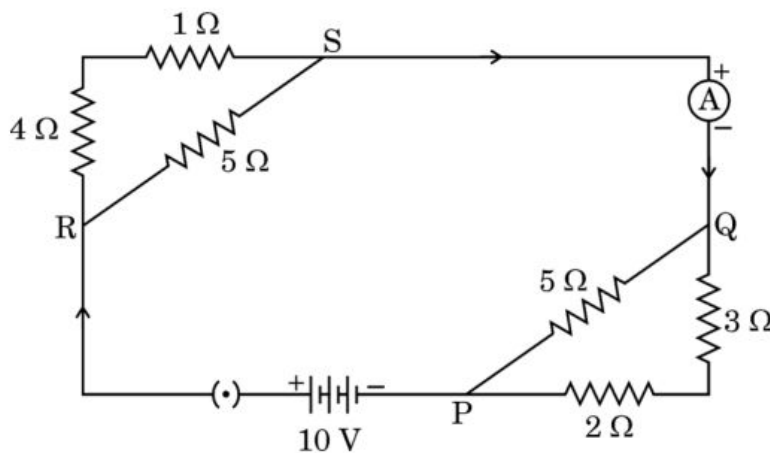
- A. We have four resistors A, B, C and D of resistance $3\ \Omega$, $6\ \Omega$, $9\ \Omega$ and $12\ \Omega$ respectively. Find out the lowest resistance which can be obtained by combining these four resistors.

Source/editorial note: Official MS arithmetic error retained: $1/12 = 3/36$, not $2/36$. Independently calculated correct minimum = $36/25 = 1.44\ \Omega$; this correction is not substituted into the official answer. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence. Official answer unavailable/withheld for an identified error, omission or incomplete verification: Official arithmetic answer $1.5\ \text{ohm}$ is erroneous; correct $1.44\ \text{ohm}$ documented separately. Consult the linked original marking scheme with this limitation in mind.

Section C — 3-mark questions

9. [3 marks] Source Q37 • 2026 First Board

37. Consider the given electric circuit :



Calculate the following :

3

- (a) Total resistance of the circuit
- (b) The electric current drawn from the battery
- (c) Potential difference between points P and Q

31/3/1

Page 27 of 31

P.T.O.

Source/editorial note: Question text, numbering, printed marks, subparts/internal choices, diagram labels and complete crop boundaries visually checked against saved original-PDF replacement crops, using JPEG quality 80 and maximum dimension 900 px. Whitespace normalized; original wording retained. No MCQ options in this question. Page furniture is not question text. Answer and marking scheme not verified. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

10. [3 marks] Source Q13 • main

- (a) List the factors on which the resistance of a conductor in the shape of a wire depends.
- (b) Why are metals good conductors of electricity whereas glass is a bad conductor of electricity ? Give reason.
- (c) Why are alloys commonly used in electrical heating devices ? Give reason.

Source/editorial note: Occurrence mappings retained verbatim; only primary PDF/question evidence is accepted and counted. Other occurrences require their own evidence validation. Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

11. [3 marks] Source Q10 • main

10. (a) Calculate the resistance of a metal wire of length 2 m and area of cross-section $1.55 \times 10^{-6} \text{ m}^2$.

(Resistivity of the metal is $2.8 \times 10^{-8} \Omega \text{ m}$)

- (b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices ?

2+1

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

12. [3 marks] Source Q32 • 2024-25

32

- A. State the relationship between the resistance R of a wire to its length l and cross sectional area A . Use the mathematical symbols to arrive at the final formula.
- B. Using the formula define the resistivity of a material.

3

Source/editorial note: The intermediate MS proportionality is printed $R \propto l/A$; retained rather than silently changing it to $R \propto 1/A$. Source-verified against complete local text/coordinates and question crops; diagram-dependent and image-only formula evidence visually checked. Layout whitespace and mathematical glyphs normalized; wording/known source errors retained. Answer text is a source-faithful linear transcription; original layout is in answer evidence.

Section D — 5-mark / case-based questions

13. [4 marks] Source Q39 • main

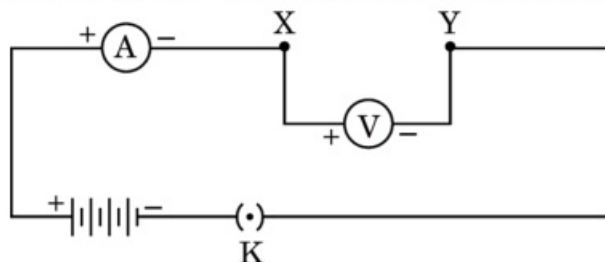
- 39.** As shown in the diagram, an electric circuit consisting of an ammeter, a voltmeter, 4 cells of 1.5 V each, a plug key with a gap XY was set up. Voltmeter and ammeter readings were recorded in the observation table for four arrangements as given below :

Arrangement No. 1 – only resistor R_1 in gap XY

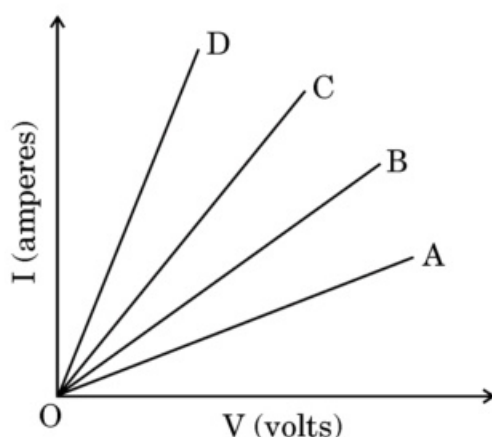
Arrangement No. 2 – only resistor R_2 in gap XY

Arrangement No. 3 – Resistors R_1 and R_2 in series in gap XY

Arrangement No. 4 – Resistors R_1 and R_2 in parallel in gap XY



Based on the observations, four V – I graphs A, B, C and D as shown in figure were drawn. Study these graphs.



- (a) Which one of the graphs represents the series combination of R_1 and R_2 ? 1
- (b) Which one of these graphs represents the parallel combination of R_1 and R_2 ? 1
- (c) (i) Show an arrangement of three resistors, each of resistance $10\ \Omega$, so that the combination has a resistance of $15\ \Omega$. Give justification for your answer. 2

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

14. [5 marks] Source Q36 • compartment

36. (a) (i) Write the relation between the electrical resistivity ' ρ ' and the resistance ' R ' of a cylindrical conductor of length ' l ' and area of cross-section ' A '. Use this relationship to determine the SI unit of ' ρ '.

(ii) Find the electrical resistivity of the material of a wire of length 120 cm and diameter 0.4 mm. The resistance of this wire is 3 Ω .

5

Source/editorial note: Official answer unavailable/withheld for an identified error, omission or incomplete verification: No independently verified official answer supplied. Consult the linked original marking scheme with this limitation in mind.

Solutions and verification status

Only explicitly verified solutions are printed. A missing solution is not a claim that the question is unsolvable.

Question 1 — pyq-2024-a415e02467405f8b9fbacfc3

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 2 — cbse_sqp_2019-20_electricity_q6

ii) $40\ \Omega$. $V = IR$; $V = 4\text{ V}$; $I = 100\text{ mA} = 0.1\text{ A}$. $R = V/I = 4/0.1\ \Omega = 40\ \Omega$.

6	ii) = $40\ \Omega$. $V=IR$, $V = 4\text{ V}$, $I = 100\text{ mA} = 0.1\text{ A}$ Hence $R = V/I = 4/0.1\ \Omega = 40\ \Omega$.	1 mark
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Marking-scheme PDF page(s): [1]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_MS.pdf

Question 3 — sqp_2022-23_q13

(c) increases

13.	(c) increases	1
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Marking-scheme PDF page(s): [1]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-MS.pdf

Question 4 — pyq-2023-dc74f0e84e53633b06097022

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 5 — SQP_2023-24_q25_main

Resistance of each part is $R/3\ \Omega$ (as resistance is proportional to the length of the wire.) [0.5]

$$1/R_1 = 3/R + 3/R + 3/R = 9/R \quad [1]$$

$$\therefore R_1 = R/9 \therefore R_1/R = 1/9 \quad [0.5]$$

25	Resistance of each part is $\frac{R}{3}\ \Omega$ (as resistance is proportional to the length of the wire.) $\frac{1}{R_1} = \frac{3}{R} + \frac{3}{R} + \frac{3}{R} = \frac{9}{R}$ $\therefore R_1 = \frac{R}{9} \therefore \frac{R_1}{R} = \frac{1}{9}$ OR	2 [0.5] [1] [0.5]
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Marking-scheme PDF page(s): [2]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-MS.pdf

Question 6 — SQP_2025-26_q34_VI_A

(i) Maximum Resistance:

To get the maximum resistance, connect all four resistors in series.

The total resistance $R_{\max}$ in series is the sum of the individual resistance:

$$R_{\max} = R + R + R + R = 4R$$

(ii) Minimum Resistance:

To get the minimum resistance, connect all four resistors in parallel.

The total resistance $R_{\min}$ in parallel is given by:

$$1/R_{\min} = 1/R + 1/R + 1/R + 1/R = 4/R \Rightarrow R_{\min} = R/4$$

For visually impaired students

A.

(i) Maximum Resistance:

- To get the maximum resistance, connect all four resistors in series.
- The total resistance $R_{\max}$ in series is the sum of the individual resistance:

$$R_{\max} = R + R + R + R = 4R$$

(ii) Minimum Resistance:

- To get the minimum resistance, connect all four resistors in parallel.

- The total resistance $R_{\min}$ in parallel is given by:

$$\frac{1}{R_{\min}} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{4}{R}$$

$$R_{\min} = \frac{R}{4}$$

Marking-scheme PDF page(s): [7, 8]

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-MS.pdf

Question 7 — SQP_2026-27_q34_B

B.

I. Definition: The resistivity (ρ) of a material is defined as the resistance of a wire having unit cross-sectional area and unit length

$R = \rho L/A$, where R is resistance of a wire of length L and cross-sectional area A (1)

II. Intrinsic property justification: Resistivity depends only on material (not on dimensions).

Example: Two copper wires of different lengths and thick-nesses yield the same ρ when computed using $R \cdot A/L$. (1)

B.

I. Definition: The resistivity (ρ) of a material is defined as the resistance of a wire having unit cross-sectional area and unit length

$R = \frac{\rho L}{A}$, where R is resistance of a wire of length L and cross-sectional area A (1)

II. Intrinsic property justification: Resistivity depends only on material (not on dimensions). Example: Two copper wires of different lengths and thick-nesses yield the same ρ when computed using $R \cdot A/L$. (1)

Question 8 — SQP_2024-25_q25_VI_A

Official answer unavailable/withheld: Official arithmetic answer 1.5 ohm is erroneous; correct 1.44 ohm documented separately.

Original marking scheme (see limitation above): https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-MS.pdf

Question 9 — pyq-2026-8f00f73d0689321485906fcf

Official answer unavailable/withheld: No independently verified official answer supplied.

Original marking scheme (see limitation above): https://www.cbse.gov.in/cbsenew/marketing-scheme/2026/X/086_SCIENCE.zip

Question 10 — pyq-2018-910a1f81e814c508e7ea459d

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 11 — pyq-2022-cfec2ef8067dd02793a5ebf2

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 12 — SQP_2024-25_q32

A. $R \propto l$

$R \propto l/A$

$R = \rho l/A$. (1.5)

B. $R = \rho l/A$

$\rho = R A/l$ for $A = 1\text{m}^2$ and $l = 1\text{m}$, we have

$\rho = R$

Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of 1 m^2 . (1.5)

32	A. $R \propto l$ $R \propto \frac{l}{A}$ $R = \rho \frac{l}{A}$. (1.5) B. $R = \rho \frac{l}{A}$ $\rho = R \frac{A}{l}$ for $A = 1\text{m}^2$ and $l = 1\text{m}$, we have $\rho = R$ Hence resistivity is the resistance offered by a wire of length 1 m having a cross sectional area of 1 m^2. (1.5)	3
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Question 13 — pyq-2025-3c907fffc20ec05c468bbbea

Official answer unavailable/withheld: No independently verified official answer supplied.

Question 14 — pyq-2024-17a293a3889e84f2b1aa8c67

Official answer unavailable/withheld: No independently verified official answer supplied.

Per-question source appendix

Original CBSE questions/images remain attributed to their source. Compilation selection and numbering are editorial. The full Science source PDFs may include unrelated chapters; only selected Electricity crops form this worksheet.

1. pyq-2024-a415e02467405f8b9fbacfc3

CBSE — Class X Science — 2024 — main — 31/4/1 | PYQ | Year/session: 2024 / main | Paper code: 31/4/1 | Source question: 14 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 7

<https://www.cbse.gov.in/cbsenew/question-paper/2024/X/SCIENCE.zip>

Also used in: official-02, official-07, official-09, official-12, official-15

2. cbse_sqp_2019-20_electricity_q6

Class-X Science-086 SAMPLE QUESTION PAPER 2019-20 | SQP | Year/session: 2019-20 / 2019-20 | Paper code: Not recorded | Source question: 6 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2019_20/Science_SQP.pdf

Also used in: official-01, official-07, official-10, official-15

3. sqp_2022-23_q13

CBSE Class X Science (086) Sample Question Paper 2022-23 | SQP | Year/session: 2022-23 / 2022-23 | Paper code: Not recorded | Source question: 13 | Original section: Section A | Selected marks: 1 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2022_23/Science-SQP.pdf

Also used in: official-03, official-05, official-09, official-13, official-15

4. pyq-2023-dc74f0e84e53633b06097022

CBSE — Class X Science — 2023 — compartment | PYQ | Year/session: 2023 / compartment | Paper code: Not recorded | Source question: 15 | Original section: SECTION A | Selected marks: 1 | PDF pages (one-based): 11

<https://www.cbse.gov.in/cbsenew/question-paper/2023-COMPTT/X/Science.zip>

Also used in: official-02, official-03, official-09, official-11, official-14

5. SQP_2023-24_q25_main

CBSE Class X Science (086) Sample Question Paper 2023-24 (English) | SQP | Year/session: 2023-24 / 2023-24 | Paper code: Not recorded | Source question: 25 main | Original section: Section-B | Selected marks: 2 | PDF pages (one-based): 4

https://cbseacademic.nic.in/web_material/SQP/ClassX_2023_24/Science-SQP.pdf

Also used in: official-05

6. SQP_2025-26_q34_VI_A

CBSE Class X Science (086) Sample Question Paper 2025-26 (English) | SQP | Year/session: 2025-26 / 2025-26 | Paper code: Not recorded | Source question: 34 VI A | Original section: Section – C | Selected marks: 2 | PDF pages (one-based): 12

https://cbseacademic.nic.in/web_material/SQP/ClassX_2025_26/Science-SQP.pdf

Also used in: official-03, official-11

7. SQP_2026-27_q34_B

CBSE Class X Science (086) Sample Question Paper 2026-27 (English) | SQP | Year/session: 2026-27 / 2026-27 | Paper code: Not recorded | Source question: 34 B | Original section: Section – C | Selected marks: 2 | PDF pages (one-based): 12

https://cbseacademic.nic.in/web_material/SQP/ClassX_2026_27/Science-SQP.pdf

Also used in: official-02, official-14

8. SQP_2024-25_q25_VI_A

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 25 VI A | Original section: Section-B | Selected marks: 2 | PDF pages (one-based): 7

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Also used in: official-05, official-12

9. pyq-2026-8f00f73d0689321485906fcf

CBSE Class X Science 2026 First Board 31-3-1 | PYQ | Year/session: 2026 / 2026 First Board | Paper code: 31/3/1 | Source question: 37 | Original section: Section C (Physics) | Selected marks: 3 | PDF pages (one-based): 27

<https://www.cbse.gov.in/cbsenew/question-paper/2026/X/Science.zip>

No cross-paper reuse.

10. pyq-2018-910a1f81e814c508e7ea459d

CBSE — Class X Science — 2018 — 31/1 — main | PYQ | Year/session: 2018 / main | Paper code: 31/1 | Source question: 13 | Original section: Section - A | Selected marks: 3 | PDF pages (one-based): 6

https://www.cbse.gov.in/curric~1/qpms2018/qp_10/Science.zip

Also used in: official-01, official-10

11. pyq-2022-cfec2ef8067dd02793a5ebf2

CBSE — Class X Science — 2022 — main — 31/1/2 | PYQ | Year/session: 2022 / main | Paper code: 31/1/2 | Source question: 10 | Original section: SECTION B | Selected marks: 3 | PDF pages (one-based): 7

<https://www.cbse.gov.in/cbsenew/question-paper/2022/X/Scince.zip>

Also used in: official-04, official-12

12. SQP_2024-25_q32

CBSE Class X Science (086) Sample Question Paper 2024-25 (English) | SQP | Year/session: 2024-25 / 2024-25 | Paper code: Not recorded | Source question: 32 | Original section: Section-C | Selected marks: 3 | PDF pages (one-based): 9

https://cbseacademic.nic.in/web_material/SQP/ClassX_2024_25/Science-SQP.pdf

Also used in: official-11, official-15

13. pyq-2025-3c907fffc20ec05c468bbbea

CBSE — Class X Science — 2025 — main — 31/3/1 | PYQ | Year/session: 2025 / main | Paper code: 31/3/1 | Source question: 39 | Original section: SECTION E | Selected marks: 4 | PDF pages (one-based): 25, 27

https://www.cbse.gov.in/cbsenew/question-paper/2025/X/086_Science.zip

Also used in: official-09, official-14

14. pyq-2024-17a293a3889e84f2b1aa8c67

CBSE — Class X Science — 2024 — compartment — 31/S/1 | PYQ | Year/session: 2024 / compartment | Paper code: 31/S/1 | Source question: 36 | Original section: SECTION D | Selected marks: 5 | PDF pages (one-based): 23

<https://www.cbse.gov.in/cbsenew/question-paper/2024-COMPTT/X/SCIENCE.zip>

Also used in: official-03, official-04, official-13, official-15